1.

Exercise 5

Given the 3 key-value pairs $\{(\boldsymbol{k}_1, \boldsymbol{v}_1), (\boldsymbol{k}_2, \boldsymbol{v}_2), (\boldsymbol{k}_3, \boldsymbol{v}_3)\} = \{([2,0,1]^\top, [-1,1]^\top), ([0,1,0]^\top, [-1,0]^\top), ([0,2,0]^\top, [0,1]^\top)\}$, and the query $\boldsymbol{q} = [1,2,3]^\top$, compute the output of attention pooling for additive attention, and scale dot-product attention, respectively. The necessary weights are generated randomly having values from the set $\{-1,0,1\}$ (you can generate them yourself, not all zeros).

Solution:

- we have the following formulas:

$$f(\boldsymbol{q}, (\boldsymbol{k}_1, \boldsymbol{v}_1), \ldots, (\boldsymbol{k}_m, \boldsymbol{v}_m)) = \sum_{i=1}^{m} \alpha(\boldsymbol{q}, \boldsymbol{k}_i)\boldsymbol{v}_i \in \mathbb{R}^v,$$

$$\alpha(\boldsymbol{q}, \boldsymbol{k}_i) = \text{softmax}(a(\boldsymbol{q}, \boldsymbol{k}_i)) = \frac{\exp(a(\boldsymbol{q}, \boldsymbol{k}_i))}{\sum_{j=1}^{m} \exp(a(\boldsymbol{q}, \boldsymbol{k}_j))} \in \mathbb{R},$$

where the query $\boldsymbol{q} \in \mathbb{R}^q$, the m key-value pairs $(\boldsymbol{k}_1, \boldsymbol{v}_1), \ldots, (\boldsymbol{k}_m, \boldsymbol{v}_m)$, where any $\boldsymbol{k}_i \in \mathbb{R}^k$ and any $\boldsymbol{v}_i \in \mathbb{R}^v$

- *additive attention*:

$$a(\boldsymbol{q}, \boldsymbol{k}) = \boldsymbol{w}_v^\top \tanh(\boldsymbol{W}_q\boldsymbol{q} + \boldsymbol{W}_k\boldsymbol{k}) \in \mathbb{R},$$

where the parameters are $\boldsymbol{W}_q \in \mathbb{R}^{h \times q}$, $\boldsymbol{W}_k \in \mathbb{R}^{h \times k}$, and $\boldsymbol{w}_v \in \mathbb{R}^h$

- *scaled dot-product attention*:

$$a(\boldsymbol{q}, \boldsymbol{k}) = \boldsymbol{q}^\top \boldsymbol{k}/\sqrt{d} \in \mathbb{R},$$

where $q = k = d$

- in our case, $m = 3$, $k = 3$, $v = 2$, $q = 3$
- we take $h = 2$ and

↑ ales

$$\boldsymbol{w}_v = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \in \mathbb{R}^2,$$

$$\boldsymbol{W}_q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \in \mathbb{R}^{2\times 3},$$

$$\boldsymbol{W}_k = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \in \mathbb{R}^{2\times 3}.$$

$m = 3$ (3 pairs)
$k = 3$ ($k_i \to$ 3 coloane)
$v = 2$ ($v_i \to$ 2 coloane)
$q = 3$ ($q \to$ 3 coloane)

- we apply the formula for additive attention:

$$\begin{aligned} a(\boldsymbol{q}, \boldsymbol{k}_1) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh\left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \tanh 4 \\ \tanh 2 \end{bmatrix} \\ &= \tanh 4 \in \mathbb{R}, \end{aligned}$$

$$\begin{aligned} a(\boldsymbol{q}, \boldsymbol{k}_2) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh\left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \tanh 5 \\ \tanh 2 \end{bmatrix} \\ &= \tanh 5 \in \mathbb{R}, \end{aligned}$$

$$\begin{aligned} a(\boldsymbol{q}, \boldsymbol{k}_3) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh\left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \tanh 6 \\ \tanh 2 \end{bmatrix} \\ &= \tanh 6 \in \mathbb{R}, \end{aligned}$$

$$
\begin{aligned}
f(\boldsymbol{q},(\boldsymbol{k}_1,\boldsymbol{v}_1),(\boldsymbol{k}_2,\boldsymbol{v}_2),(\boldsymbol{k}_3,\boldsymbol{v}_3)) &= \frac{\exp(\tanh 4)}{\exp(\tanh 4)+\exp(\tanh 5)+\exp(\tanh 6)}\begin{bmatrix}-1\\1\end{bmatrix} \\
&\quad+\frac{\exp(\tanh 5)}{\exp(\tanh 4)+\exp(\tanh 5)+\exp(\tanh 6)}\begin{bmatrix}-1\\0\end{bmatrix} \\
&\quad+\frac{\exp(\tanh 6)}{\exp(\tanh 4)+\exp(\tanh 5)+\exp(\tanh 6)}\begin{bmatrix}0\\1\end{bmatrix} \\
&= \frac{1}{\exp(\tanh 4)+\exp(\tanh 5)+\exp(\tanh 6)} \\
&\quad\times\begin{bmatrix}-\exp(\tanh 4)-\exp(\tanh 5)\\ \exp(\tanh 4)+\exp(\tanh 6)\end{bmatrix}\in\mathbb{R}^2.
\end{aligned}
$$

now, we apply the formula for scaled dot-product attention:

$$
a(\boldsymbol{q},\boldsymbol{k}_1)=\frac{1}{\sqrt{3}}\begin{bmatrix}1&2&3\end{bmatrix}\begin{bmatrix}2\\0\\1\end{bmatrix}=\frac{5}{\sqrt{3}}\in\mathbb{R},
$$

$$
a(\boldsymbol{q},\boldsymbol{k}_2)=\frac{1}{\sqrt{3}}\begin{bmatrix}1&2&3\end{bmatrix}\begin{bmatrix}0\\1\\0\end{bmatrix}=\frac{2}{\sqrt{3}}\in\mathbb{R},
$$

$$
a(\boldsymbol{q},\boldsymbol{k}_3)=\frac{1}{\sqrt{3}}\begin{bmatrix}1&2&3\end{bmatrix}\begin{bmatrix}0\\2\\0\end{bmatrix}=\frac{4}{\sqrt{3}}\in\mathbb{R},
$$

$$
\begin{aligned}
f(\boldsymbol{q},(\boldsymbol{k}_1,\boldsymbol{v}_1),(\boldsymbol{k}_2,\boldsymbol{v}_2),(\boldsymbol{k}_3,\boldsymbol{v}_3)) &= \frac{\exp\left(\frac{5}{\sqrt{3}}\right)}{\exp\left(\frac{5}{\sqrt{3}}\right)+\exp\left(\frac{2}{\sqrt{3}}\right)+\exp\left(\frac{4}{\sqrt{3}}\right)}\begin{bmatrix}-1\\1\end{bmatrix} \\
&\quad+\frac{\exp\left(\frac{2}{\sqrt{3}}\right)}{\exp\left(\frac{5}{\sqrt{3}}\right)+\exp\left(\frac{2}{\sqrt{3}}\right)+\exp\left(\frac{4}{\sqrt{3}}\right)}\begin{bmatrix}-1\\0\end{bmatrix} \\
&\quad+\frac{\exp\left(\frac{4}{\sqrt{3}}\right)}{\exp\left(\frac{5}{\sqrt{3}}\right)+\exp\left(\frac{2}{\sqrt{3}}\right)+\exp\left(\frac{4}{\sqrt{3}}\right)}\begin{bmatrix}0\\1\end{bmatrix} \\
&= \frac{1}{\exp\left(\frac{5}{\sqrt{3}}\right)+\exp\left(\frac{2}{\sqrt{3}}\right)+\exp\left(\frac{4}{\sqrt{3}}\right)} \\
&\quad\times\begin{bmatrix}-\exp\left(\frac{5}{\sqrt{3}}\right)-\exp\left(\frac{2}{\sqrt{3}}\right)\\ \exp\left(\frac{5}{\sqrt{3}}\right)+\exp\left(\frac{4}{\sqrt{3}}\right)\end{bmatrix}\in\mathbb{R}^2.
\end{aligned}
$$

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② Exercitiu curs BIA S11

$k_1 = \begin{pmatrix}-1\\2\end{pmatrix}$ $k_2 = \begin{pmatrix}1\\-1\end{pmatrix}$ $k_3 = \begin{pmatrix}2\\0\end{pmatrix}$

$v_1 = \begin{pmatrix}-1\\0\\1\end{pmatrix}$ $v_2 = \begin{pmatrix}1\\1\\-2\end{pmatrix}$ $v_3 = \begin{pmatrix}0\\1\\-1\end{pmatrix}$

$q = \begin{pmatrix}2\\3\end{pmatrix}$

$k_i \in \mathbb{R}^{2\times 1}$ $v_i \in \mathbb{R}^{3\times 1}$

$q \in \mathbb{R}^{2\times 1}$

$m = 3$, $k = 2$, $v = 3$, $q = 2$, $h = 2$

$W_v \in \mathbb{R}^{2=h} \Rightarrow W_v = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$W_q \in \mathbb{R}^{h \times q = 2 \times 2} \Rightarrow W_q = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$W_k \in \mathbb{R}^{h \times k = 2 \times 2} \Rightarrow W_k = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$

additive attention

$$a(q, k_1) = W_v^T \cdot \tanh(W_q q + W_k \cdot k_1) =$$

$$= (0 \quad 1) \tanh\left(\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -1 \\ 2 \end{pmatrix}\right) =$$

$$= (0,1) \tanh\left(\begin{pmatrix} 3 \\ -2 \end{pmatrix} + \begin{pmatrix} -3 \\ 2 \end{pmatrix}\right) = (0,1)\begin{pmatrix} \tanh(0) \\ \tanh(0) \end{pmatrix} =$$

$$= \tanh(0)$$

La fel și pt restul

$a(q, k_2) = \tanh(-3)$

$a(q, k_3) = \tanh(-2)$

$$f(q, (k_1, v_1), \dots, (k_3, v_3)) = \sum_{i=1}^{m} \alpha(q, k_i) v_i$$

$$\alpha(q, k_i) = \text{softmax}(a(q, k_i)) = \frac{\exp(a(q, k_i))}{\sum_{j=1}^{m} \exp(a, q, k_i)}$$

$$f = \frac{1}{\exp(\tanh(0)) + \exp(\tanh(-3)) + \exp(\tanh(-2))} \cdot$$

$$\left(\exp(\tanh(0)) \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \exp(\tanh(-3)) \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} + \exp(\tanh(-2)) \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right) =$$

$$= \frac{1}{\exp(\tanh(0)) + \exp(\tanh(-3)) + \exp(\tanh(-2))} \cdot \begin{pmatrix} -e^{\tanh(0)} + e^{\tanh(-3)} \\ e^{\tanh(-3)} + e^{\tanh(-2)} \\ e^{\tanh(0)} - 2e^{\tanh(-3)} - e^{\tanh(-2)} \end{pmatrix}$$

dot product attention

$$a(q, k_1) = \frac{1}{\sqrt{2}} \begin{pmatrix} 2 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \frac{4}{\sqrt{2}}$$

$$a(q, k_2) = -\frac{1}{\sqrt{2}}$$

$$a(q, k_3) = \frac{4}{\sqrt{2}}$$

$$f = \frac{1}{e^{\frac{4}{\sqrt{2}}} + e^{-\frac{1}{\sqrt{2}}} + e^{\frac{4}{\sqrt{2}}}} \cdot \left(e^{\frac{4}{\sqrt{2}}} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + e^{-\frac{1}{\sqrt{2}}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} + e^{\frac{4}{\sqrt{2}}} \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \right)$$

3. Compute the total number of parameters for additive attention for n queries and m key-value pairs, where queries are of length q, keys are of length k, and values are of length v. Give the answer symbolically. Numerical application: $n = 2, m = 3, q = k = 2, v = 3$.

$$a(q, k) = w_v^T \tanh(W_q q + W_k k)$$

$W_q \in \mathbb{R}^{h \times q} = \mathbb{R}^{h \times 2} \Rightarrow 2 \cdot h$

$W_k \in \mathbb{R}^{h \times k} = \mathbb{R}^{h \times 2} \Rightarrow 2 \cdot h$

$W_v \in \mathbb{R}^{h} = \mathbb{R}^{h} \Rightarrow h$

$5h$

$m = 3 \Rightarrow$ avem 3 perechi $\Rightarrow 3 \cdot 5 \cdot h = 15h$ NU

4. Question 9
Not yet answered
Marked out of 1.00
Flag question

Assume a scalar multiplication and a scalar addition both account for one FLOP and exponentiation accounts for 10 FLOPs. All other scalar operations account for 0 FLOPs. How many FLOPs does the scaled dot-product attention require for n queries and m key-value pairs, where queries and keys are of length d and values are of length v. Give the answer symbolically. Numerical application: $n = 2, m = 3, d = 2, v = 3$.

scalar multiplication → 1 FLOPS
scalar addition → 1 FLOPS
exponentiation → 10 FLOPS
other scalar operation → 0 FLOPS

scaled dot product?
$n = 2$, $m = 3$, $d = 2$, $v = 3$

$$a(q, k) = q^T \cdot k \;/\; \sqrt{d}$$

query, key, input

$Q \in \mathbb{R}^{n \times d}$, $K \in \mathbb{R}^{m \times d}$, $V \in \mathbb{R}^{m \times v}$

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \in \mathbb{R}^{n \times v}$$

pt un query și un key

$q \in \mathbb{R}^{1 \times d}$, $k \in \mathbb{R}^{1 \times d}$ $\Rightarrow$ $q^T \cdot k \in \mathbb{R}^{d \times d}$

fiecare el. se înmulțește cu fiecare el $\Rightarrow d \cdot d \cdot 1$ flops → multiplication

$(q^T \cdot k) \cdot \frac{1}{\sqrt{d}}$ $\Rightarrow$ încă $d \cdot d \cdot 1$ flops

$a(q, k) \longrightarrow 2 \cdot d^2$ flops

$Q \in \mathbb{R}^{n \times d}$

$K \in \mathbb{R}^{m \times d} \rightarrow K^T \in \mathbb{R}^{d \times m}$ $\Rightarrow$ $n \times m$

(5.)

5. Given the 4 key-value pairs $\{(\boldsymbol{k}_1, \boldsymbol{v}_1), (\boldsymbol{k}_2, \boldsymbol{v}_2), (\boldsymbol{k}_3, \boldsymbol{v}_3), (\boldsymbol{k}_4, \boldsymbol{v}_4)\} = \{([2,0]^\top, [-1,1,0]^\top), ([0,1]^\top, [-1,0,0]^\top), ([0,2]^\top, [0,1,-1]^\top), ([-1,0]^\top, [1,2,1]^\top)\}$, and the query $\boldsymbol{q} = [1,2]^\top$, compute the output of attention pooling for scaled dot-product attention.

$m = 4$ pairs

$k = 2$ coloane k_1^T altfel linii

$v = 3$ coloane v_1^T altfel linii

$q = 2$ coloane q^T altfel linii

$q = k = d = 2$

$$a(q, k) = q^T k / \sqrt{d} \in \mathbb{R}$$

$$a(q, k_1) = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$a(q, k_2) = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$a(q, k_3) = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \end{pmatrix} \frac{1}{\sqrt{2}} = \frac{4}{\sqrt{2}} = \frac{4\sqrt{2}}{2} = 2\sqrt{2}$$

$$a(q, k_4) = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \end{pmatrix} \frac{1}{\sqrt{2}} = \frac{-1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$$

$$f(q, (k_i, v_i)) = \sum_{i=1}^{m} \alpha(q, k_i) v_i \in \mathbb{R}^v$$

$$\alpha(q, k_i) = \text{softmax}(a(q, k_i)) = \frac{\exp(a(q, k_i))}{\sum_{j=1}^{m} \exp(a(q, k_i))}$$

$$f(q, (k_i, v_i)) = \frac{1}{e^{\sqrt{2}} + e^{\sqrt{2}} + e^{2\sqrt{2}} + e^{-\frac{\sqrt{2}}{2}}} \cdot$$

$$\cdot \left(e^{\sqrt{2}} \cdot \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + e^{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} + e^{2\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + e^{-\frac{\sqrt{2}}{2}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right) =$$

$$= \frac{1}{e^{\sqrt{2}} + e^{\sqrt{2}} + e^{2\sqrt{2}} + e^{-\frac{\sqrt{2}}{2}}} \begin{pmatrix} -2e^{\sqrt{2}} + e^{-\frac{\sqrt{2}}{2}} \\ e^{\sqrt{2}} + e^{2\sqrt{2}} + 2e^{-\frac{\sqrt{2}}{2}} \\ -e^{2\sqrt{2}} + e^{-\frac{\sqrt{2}}{2}} \end{pmatrix}$$

9. Given the 4 key-value pairs $\{(\boldsymbol{k}_1, \boldsymbol{v}_1), (\boldsymbol{k}_2, \boldsymbol{v}_2), (\boldsymbol{k}_3, \boldsymbol{v}_3), (\boldsymbol{k}_4, \boldsymbol{v}_4)\} = \{([2,0]^\top, [-1,1,0]^\top), ([0,1]^\top, [-1,0,0]^\top), ([0,2]^\top, [0,1,-1]^\top), ([-1,0]^\top, [1,2,1]^\top)\}$, and the query $\boldsymbol{q} = [1,2]^\top$, compute the output of attention pooling for scaled dot-product attention.

10. Define an MLP for

aceleași valori